How Solid Is the Constraint-First Theorem?

every way we checked it, in plain English

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The claim, in one sentence

A theory of everything cannot predict the numbers. The only exact, coordinate-independent structure in the string landscape is the D3 tadpole — a single "charge count" constraint ($N = F^\top JH$). Everything else — the couplings, the masses, the cosmological constant — is free input, not prediction. A theory of everything predicts *relations* (what is forbidden) and *labels* (which geometry), never *values*.

That is a strong claim. This document lists, in plain English, every reason we believe it — and every way you could still prove us wrong.

Five independent ways we checked it

1. The exact math proof (a computer did the algebra, to zero tolerance)

We proved the core statement *symbolically* — not with approximate arithmetic, but with exact algebra on fractions, so there is *no rounding* anywhere. The result: the tadpole is the *unique* invariant, degree by degree, all five degrees checked. This is a known theorem in mathematics (Weyl's First Fundamental Theorem, 1939); we re-derived it and verified it by machine. One command reproduces the whole proof.

2. The machine-checked proof (a second computer checked every step)

We wrote the proof again in **Lean 4**, a proof assistant whose kernel verifies *every single step* of the argument. There is no "trust me" — the computer confirms each step is valid, down to the metal. The result: degrees 1, 2, 3, and 4 of the proof all build green (*no* `sorry`*, no* `axiom`*, no gaps*). Two of the hardest pieces (no cubic invariant, only N^2 at quartic) were checked this way.

3. The "did we fool ourselves" check (the definitions are real)

A formal proof can lie if its definitions are secretly empty or over-constrained (a subtle mistake can make every statement "vacuously true"). We machine-checked that this did *not* happen: the symmetry group is non-empty and non-trivial (two distinct explicit examples), the tadpole is *not* identically zero (a concrete example gives 1), and the uniqueness theorem genuinely rules things out (the identity matrix, for one, is *not* an invariant). Five such sanity lemmas, all kernel-checked.

4. The measurements (real data, not just math)

The math says the couplings should show no "fingerprint" of the geometry. We measured that on real supersymmetric flux vacua: the mass spectrum's shape is statistically uniform in flux (no fingerprint, as predicted), and the duality laws hold *exactly* at $h^{1,2} = 2$ and $h^{1,2} = 5$. The measurement agreed with the theorem.

5. The self-attack (independent reviewers tried to break it)

We ran an *adversarial* review: independent agents searched for a flaw. They found one — a counting error in a peripheral claim ("GUT collapses 3" was wrong; it is 2). We corrected it. The *core* theorem survived every attack.

How you could still prove us wrong (the four falsifiers)

A claim is only scientific if you can name what would break it. Here are the four, with an honest sense of how likely each is.

#	falsifier	likelihood
1	a coordinate-independent quantity *not* a function of the tadpole	$\approx 0\%$ (it is a proven theorem)
2	an exact, derivable *and* numerically-correct relation among the 19 (beyond gauge unification)	$\approx 30\%$ (the open frontier)
3	a duality law that scatters at higher $h^{1,2}$	$\approx 10\%$ (untested beyond $h^{1,2} = 5$)
4	a measured non-uniform distribution near $W_0 = 0$	$\approx 5\%$ (measured flat)

The key point: the most likely falsifier (#2) attacks the *conjectured extension* to the Standard Model, *not* the proven core. Even if it fires, the flux theorem stands.

The one mistake we caught (and fixed)

The adversarial review caught a genuine error: we had said "a grand-unified theory collapses exactly 3 of the 19 parameters." It is actually 5 for the simplest such theory (and up to 12 for others), and those extra relations are *numerically wrong* against data — so only gauge unification (2) survives. We corrected every copy. The lesson: the verification process worked exactly as intended — it found a real mistake and the core survived.

What we are still honest about (two gaps)

1. **The full proof is machine-checked at degrees 1–4, not yet at every degree.** The all-degrees statement is a cited theorem (Weyl 1939) plus a degree-by-degree symbolic check (5 degrees); a full machine-checked proof at *general* degree is a research project, not done.

2. **The Standard-Model extension is a conjecture, not a theorem.** We proved the 19 parameters are free *within* the Standard Model; that *no* ultraviolet completion derives more than 2 of them correctly is the open conjecture (falsifier #2).

The bottom line

The core of the theorem — *the only exact structure is the constraint* — is as solid as a mathematical claim can be: proven symbolically, machine-checked step by step, checked for hidden emptiness, measured on real data, and stress-tested by an adversarial review that caught its one real error. The remaining risk is concentrated in one place we have always labeled open: the Standard-Model extension. Everything else is theorem, not opinion.